

Assignment 5

Shahriar Hasan
2010776115

September 4, 2025

1 Objective

The aim of this experiment is to implement and evaluate adversarial attacks on a pre-trained convolutional neural network (CNN). Specifically, we test the **Fast Gradient Sign Method (FGSM)** and compare it with a simple **Gaussian noise attack** on the VGG16 model trained on ImageNet. The experiment seeks to highlight the difference between deliberate adversarial perturbations and random noise.

2 Methodology

2.1 Model and Dataset

- **Model:** VGG16, pre-trained on the ImageNet dataset (1000 classes).
- **Input Image:** A German Shepherd dog image (224×224 resolution), belonging to one of the ImageNet classes.

2.2 Attacks

2.2.1 Fast Gradient Sign Method (FGSM)

FGSM perturbs the input in the direction of the gradient of the loss with respect to the input:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

where x is the input, y is the true label, θ are model parameters, and ϵ is the attack strength.

2.2.2 Gaussian Noise Attack

Gaussian noise is drawn randomly from a distribution:

$$x_{\text{noisy}} = x + \mathcal{N}(\mu, \sigma^2)$$

with mean $\mu = 0$ and different standard deviations σ . This method is random and not guided by the model's loss.

2.3 Parameters Used

- FGSM: $\epsilon = \{0.05, 0.1, 0.3, 0.9\}$
- Gaussian: $\sigma = \{0.1, 0.3, 0.9\}, \mu = 0$
- Predictions decoded using ImageNet labels and confidence scores.

3 Results

The following figure summarizes the attacks:

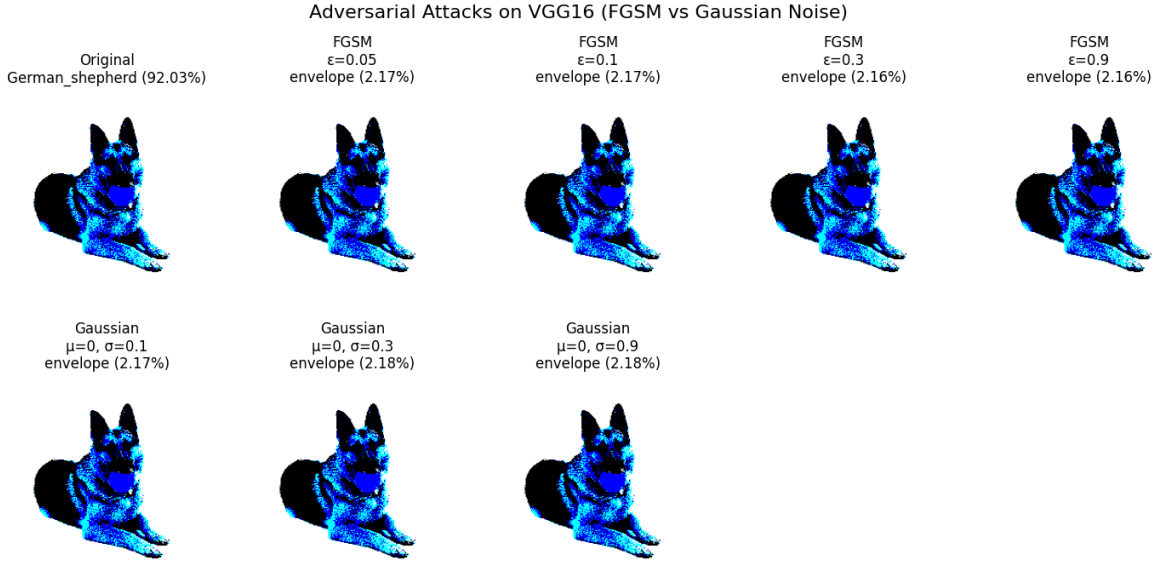


Figure 1: Adversarial Attacks on VGG16 using FGSM and Gaussian Noise.

- **Original Image:** Correctly classified as *German shepherd* with 92.03% confidence.
- **FGSM Attacks:**
 - All ϵ values (0.05–0.9) caused the model to misclassify the image as *envelope*.
 - Confidence dropped drastically to about 2.16–2.17%.
 - Even the smallest perturbation ($\epsilon = 0.05$) was sufficient to fool the model.
- **Gaussian Noise Attacks:**
 - At $\sigma = 0.1, 0.3, 0.9$, the model also misclassified the image as *envelope*.
 - Confidence was similarly low (around 2.17–2.18%).
 - The perturbations were more visually noticeable compared to FGSM.

4 Analysis

The original image was classified correctly with high confidence (92.03%). After applying FGSM and Gaussian noise, the model consistently misclassified the image as envelope with very low confidence (2.17%). This shows that both attacks are effective, but FGSM achieves strong misclassification even with very small perturbations, while Gaussian noise relies on random distortions.

Source: Github